



COMSTRUCT · ETH HACKATHON · MAY 2026

C-Flow

Tail-spend procurement,
built for the foreman.

A three-input mobile ordering app, a procurement web dashboard, and an AI catalog pipeline. End-to-end demo running live.

THE PROBLEM

Phone calls, paper notes, no trail.

On a construction site today, ordering a box of screws or a pack of safety gloves still runs on the same playbook as twenty years ago. The foreman picks up the phone, calls the merchant, takes delivery against a paper delivery note, and stacks the note on a shelf in the site container. Once a week, somebody drives a stack of those notes to headquarters, where another person matches them against invoices line by line. Errors are common. Procurement has near-zero visibility into what each site is spending on small consumables until weeks later.

The opportunity — tail spend

Procurement spend is conventionally split into A, B and C categories. A-materials (concrete, steel, windows, HVAC) are high-value and project-critical: planning teams own those decisions. C-materials are the opposite — individually cheap, but a constant stream that gums up daily operations.



Fixing C-material procurement isn't about better price negotiation. It's about removing process friction — the phone calls, the paper, the manual invoice matching, the lost time on site. That's where C-Flow plays.

THE SOLUTION

Three apps. One backbone.

C-Flow is three connected applications sitting on a shared backend:

Foreman mobile app

Three big buttons on the home screen: camera, microphone, text. The foreman picks whichever fits the moment — holding the broken bit up to the camera, speaking into the mic with gloves on, or typing on a quiet container desk. An AI classifier turns any of these into a structured request, matches it against the contracted catalog, and shows large, single-tap product cards sorted by what their site actually orders.

Procurement web dashboard

Procurement gets a live view of every site: current spend, category breakdown, top products, top suppliers, the approval queue. They set CHF caps per employee, upload supplier catalogs as Excel / PDF, and audit any order back to its requestor, approver and PO PDF.

AI backbone

A Node + Supabase backend handles classification, A-material gating, document ingestion, PO PDF generation and supplier email. Claude and OpenAI APIs do the heavy lifting on text and image understanding.

Built with: React Native (Expo) for mobile · React + Vite for web · Supabase (Postgres + Storage + Realtime) for state · Node for the API · Claude and OpenAI for AI · Resend for supplier emails.

Two taps. No SKUs.

Target persona: not a digital native. On site, often with gloves on. Time-poor. If the app takes longer than a phone call, they won't use it. C-Flow is designed for that reality.

Three input modes

- **Camera.** Hold the broken part up to the phone. AI matches shape, color and features against visual descriptions in the catalog.
- **Microphone.** Say it out loud. Whisper transcribes German, Swiss German, English. Synonyms in the catalog handle regional names.
- **Text.** A quick re-order when it's quiet. Synonyms again: 'rawl plug' and 'wall anchor' both resolve to the same SKU.

Glove-friendly UI

- Large tap targets, full-width buttons, no modal dialogs.
- No keyboard required for the camera or voice paths.
- High contrast, dust-readable typography on a phone in bright light.

Order flow

After classification, the app shows a list of matching products as large cards: name, unit price, supplier, quantity +/- buttons. One tap on **Order** sends it. The status updates in real time — pending approval, approved, PO sent, supplier confirmed, delivered — without the foreman ever having to chase anyone.

C-material in. A-material out.

The hackathon brief is explicit about this: the app must explain what C-materials are and prevent foremen from trying to order major A-materials through it. We enforce that boundary in three places at once — the database schema, the AI classifier, and the UX.

✓ ALLOWED IN C-FLOW

Small everyday items for your site.

- Fasteners — screws, anchors, nails
- Safety — gloves, helmets, glasses
- Adhesives — tape, sealant, glue
- Hand tools, cleaning, small electrical

✗ BOUNCED (A-MATERIAL)

"That's a Bauleiter decision."

- Concrete, cement, rebar, structural steel
- Windows, doors, HVAC, boilers
- Power tools (drills, saws, grinders)
- Anything > 200 CHF or requires delivery slot

How it's enforced

- **Schema.** The products.category enum is restricted to the 11 C-material categories from the canonical catalog.
- **Prompt.** The classifier returns an is_c_material boolean. False triggers a friendly redirect message.
- **UX.** First-launch onboarding card and a persistent home-screen reminder: "Small everyday items only — big stuff → Bauleiter."
- **Demo moment.** Searching for 'concrete' or 'Bohrhammer' bounces with helpful copy instead of returning a wrong match.

Smart matching, not a 50,000-SKU dump.

The risk with C-materials is dumping tens of thousands of SKUs on the foreman and calling it a feature. C-Flow's catalog is designed so that the foreman never sees the full SKU surface — the right product floats to the top.

Synonyms and visual descriptions

Every SKU in the live catalog carries a 'visual description' column (written for vision models — shape, color, distinguishing features) and a 'synonyms' column (regional and trade names). A photo of a wall plug matches against the visual description; typing 'rawl plug' matches the synonym; the same SKU returns from either path.

Popularity sort

Each completed order bumps the SKU's popularity score. The first product the foreman sees is the one their site actually orders most, not whichever SKU was registered first.

Normalize at ingest

Suppliers' categories are inconsistent — one labels a 4×40 Torx as 'fasteners > wood screw', another as 'screws / chipboard'. C-Flow collapses all of that into 11 canonical categories during catalog upload, so the foreman's search stays fast and the procurement dashboard rolls up cleanly.

Live in the demo: 91 SKUs across 11 categories — Cleaning, Fasteners, Hand Tools, Office, Adhesives, Safety, Electrical, Painting, Plumbing, Chemicals, Storage. Every row has full visual and synonym metadata for AI matching.

Control without micromanaging.

Procurement teams don't want to approve every CHF-3 box of screws. They want to set policy once and let the system enforce it. C-Flow gives them three primitives:

1. Per-employee budget caps

Each user has a CHF spending limit. Orders under the cap auto-approve and proceed directly to PO generation. Orders over the cap land in the approver's queue with full context — requestor, site, product, price, justification — and can be approved or rejected in seconds from the mobile app.

2. Live spend dashboard

A card per site with current spend vs. budget, drilling into category breakdown, top suppliers and recent orders. No more weekly delivery-note transfers from container to headquarters — the data is there the moment the supplier confirms.

3. Audit trail

Every order keeps the full chain: requestor, approver, timestamps, AI classification output, PO PDF, supplier confirmation. Useful for compliance, ESG reporting, and the inevitable "who ordered this?" conversation.

Approval logic

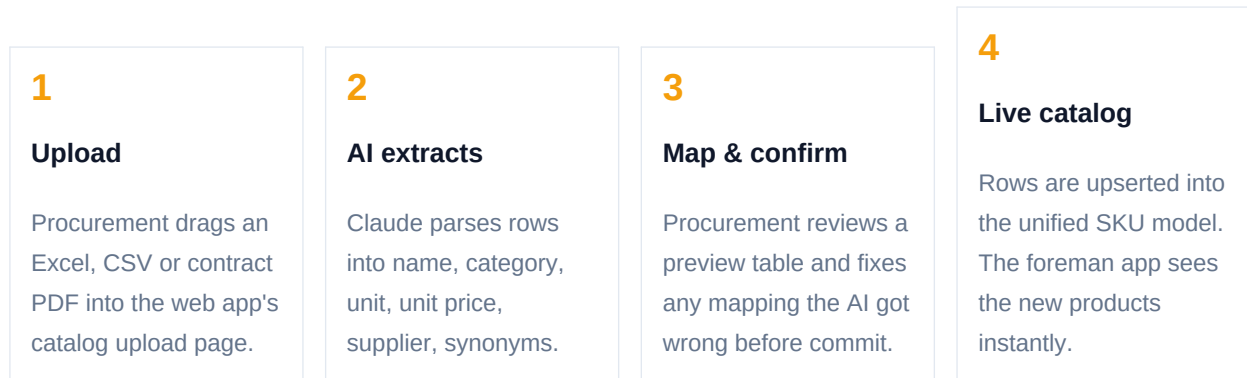
$\leq$ cap → auto-approved

$>$ cap → approver notified

A-material → blocked

Messy supplier files → clean catalog.

C-Flow's catalog isn't hand-curated. Procurement uploads supplier price lists, framework contracts and quotes as Excel, CSV or PDF — exactly how they arrive in the inbox today — and the pipeline turns them into a unified, searchable catalog the foreman app can use immediately.



Why this matters for comstruct

Onboarding a new supplier is the slowest part of any procurement platform. By accepting whatever format the supplier sends and letting AI do the mapping, C-Flow drops new-supplier setup from days to minutes. The same pipeline handles a one-off quote and a 500-row framework contract.

Proof: A 91-item C-material catalog was ingested end-to-end for the demo in roughly 30 seconds — 11 categories, visual descriptions, synonyms, units and CHF pricing all preserved through the extraction.

DEMO FLOW

How it runs end to end.

The live demo follows the seven-step flow construct provided, with no mocked steps:

1. Procurement curates

Catalog uploaded via the web app. Per-site assortments and per-employee CHF caps are set.

2. Foreman orders

Camera, voice or text. AI classifies, returns a ranked list, foreman taps Order.

3. Auto-route

Below cap → auto-approved. Above cap → queued.
A-material → blocked with a friendly redirect.

4. Approver decides

Approver sees requestor, site, product, price and AI reasoning on their mobile app. Approves or rejects in seconds.

5. PO to supplier

Backend generates a signed PO PDF, uploads it to storage, emails the supplier with a confirm link.

6. Supplier confirms

One-click confirm link flips the order to confirmed and notifies the foreman.

7. Delivered

Status closes the loop in the foreman app and the procurement dashboard updates spend per site and category.

Run time end-to-end: under four minutes in the live demo, including the deliberate A-material rejection moment.

Mapped to the four judging criteria.

USER FIT — FOREMAN

Easier than a phone call

Camera, voice and text inputs. AI matches the request to a SKU. Two taps to order, glove-friendly buttons, regional synonyms baked into the catalog.

USER FIT — PROCUREMENT

Control without micromanaging

Per-employee CHF cap auto-approves small orders. Larger orders queue with full context. Live spend dashboard by site and category. Audit trail on every order.

HANDLING OF COMPLEXITY

91 SKUs, smart matching

Each SKU carries visual descriptions for image matching and synonyms for text. Popularity score promotes what the site actually buys. 11 canonical categories at ingest.

INTEGRATION THINKING

Excel / PDF in seconds

Drag-and-drop ingestion through the procurement web app. Claude extracts rows to a unified product model. Procurement confirms the mapping. Live in the foreman app immediately.

In and out of scope

In scope (built)

- ✓ Mobile ordering app with camera, voice and text inputs
- ✓ AI classification with A-material guardrail
- ✓ Approval queue, per-employee budget caps
- ✓ Procurement web dashboard with live spend per site and category
- ✓ Catalog ingestion from CSV / XLSX / PDF (AI-extracted)
- ✓ Auto PO PDF generation and supplier email with confirmation link
- ✓ Full audit trail on every order

Out of scope (next phase)

- Native ERP / SAP integration (next phase)
- On-site inventory tracking
- Supplier-side portal app
- Multi-language UI (German fallback only)
- Offline mode without internet

Repository: github.com/anshsin7/constructionAI

Stack: Expo + React Native · React + Vite · Supabase · Node · Claude / OpenAI · Resend